

YUEZHAO LI

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PERSONALITY

Nationality	Chinese
Date of birth	18 April, 1996
Languages	Chinese (Native), English (C1), German (B1), Dutch (A1)

RESEARCH INTEREST

- ◇ Noncommutative geometry via spectral triples and unbounded Kasparov modules.
- ◇ Spectral localisers and computational (bivariant) K-theory of C^* -algebras.
- ◇ Topological phases of matter via groupoids and coarse geometry.

APPOINTMENTS

Max-Planck-Institut für Mathematik Mar 2026 — Aug 2026
Visiting postdoctoral researcher Germany
 ◇ Host: Prof. Dr. Matthias Lesch.

EDUCATION

Universiteit Leiden Oct 2021 — Feb 2026
PhD in mathematics The Netherlands
 ◇ Supervisor: Dr. Bram Mesland.
 ◇ Thesis: *Spectral localisers and aperiodic topological phases in noncommutative geometry.*

Georg-August-Universität Göttingen Oct 2018 — Aug 2021
Master in mathematics, with minor in physics Germany
 ♦ Thesis advisor: Prof. Dr. Ralf Meyer.
 ♦ Thesis: *Invariants for topological insulators coming from decompositions of coarse spaces.*

Peking University Sep 2013 — Jul 2018
Bachelor in physics, with minor in mathematics
China

- ◇ Thesis advisor: Prof. Dr. Limei Xu.
- ◇ Thesis: *Phase transition and critical phenomena.*

PUBLICATIONS

Peer-reviewed journal articles

- [1] Y. Li and B. Mesland. *The odd spectral localiser via asymptotic morphisms and quasi-projections*. Ann. K-Theory **11**, no. 2 (2026), 213–260. doi: [10.2140/akt.2026.11.213](https://doi.org/10.2140/akt.2026.11.213).
- [2] Y. Li. *Robustness of Topological Phases on Aperiodic Lattices*. Math. Phys. Anal. Geom. **29**, no. 2 (2026), Paper No. 16. doi: [10.1007/s11040-026-09547-1](https://doi.org/10.1007/s11040-026-09547-1).

Preprints

- [3] Y. Li and G. C. Thiang. *Consistent symmetry breaking and topological phases* (2026). [arXiv: 2607.28181](https://arxiv.org/abs/2607.28181).

TALKS

Research talks

- ◇ JULY 2026: *The spectral localiser via E-theory*.
Frontiers in Noncommutative Geometry and its Applications. Hangzhou, China.
- ◇ MAY 2026: *Noncommutative geometry of topological phases of matter*.
Higher Differential Geometry Seminar. Max-Planck-Institut für Mathematik, Germany.
- ◇ APRIL 2026: *The spectral localiser: what and why*.
Oberseminar Global Analysis and Operator Algebras. Universität Bonn, Germany.
- ◇ DECEMBER 2025: *The spectral localiser via E-theory*.
NCG Colloquium II. Nijmegen, Netherlands.
- ◇ NOVEMBER 2025: *The spectral localiser via E-theory*.
Oberseminar Global Analysis and Operator Algebras. Universität Bonn, Germany.
- ◇ AUGUST 2025: *On topological phases of aperiodic matter*.
NCG Day. Leiden, Netherlands.
- ◇ JULY 2025: *The spectral localiser via E-theory*.
Oberseminar Analysis. Leibniz Universität Hannover, Germany.
- ◇ JULY 2025: *The comparison maps between groupoid and Roe C^* -algebras from aperiodic topological insulators*.
Oberseminar Noncommutative Geometry. Georg-August-Universität Göttingen, Germany.
- ◇ JUNE 2025: *The spectral localiser via E-theory*.
 C^* -algebras, Coarse Geometry and Physics. Greifswald, Germany.
- ◇ APRIL 2025: *On robustness of topological phases* (short talk).
Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Space-time. Marseille, France.
- ◇ MARCH 2025: *On robustness of topological phases of matter*.
Seminar in Analysis and Applications. Delft University of Technology, Netherlands.
- ◇ JANUARY 2025: *Understanding strong and weak topological phases*.
Forschungsseminar Algebra/Geometrie/Topologie. Universität Greifswald, Germany.
- ◇ SEPTEMBER 2024: *Index theory of spectral truncations and the localisation algebra*.
East China Normal University, Shanghai, China.
- ◇ OCTOBER 2023: *Models of aperiodic topological insulators* (short talk).
Autumn School on Large-scale Geometry. Göttingen, Germany.

Colloquium talks

- ◇ JULY 2026: *K-theory in solid-state physics: modelling and computation*.
MPI-Oberseminar, Bonn, Germany.
- ◇ SEPTEMBER 2024: *Noncommutative geometry of integer quantum Hall effect*.
Sichuan University, Chengdu, China.
- ◇ MAY 2024: *A hitchhiker's guide to topological insulators in noncommutative geometry*.
Leiden–Birmingham meeting, Leiden, Netherlands.
- ◇ NOVEMBER 2023: *Noncommutative geometry in index theory and physics*.
Leiden–Luxembourg PhD Away Day. Leiden, Netherlands.

CONFERENCES, WORKSHOPS AND SEMINARS

Participation

- ◇ JULY 2026: Summer School and International Workshop on Noncommutative Geometry. Dalian, China.
- ◇ JULY 2026: Frontiers in Noncommutative Geometry and its Applications. Hangzhou, China.
- ◇ JUNE 2026: Interfaces of Spectral Theory, Operator Algebras and Noncommutative Geometry, on the occasion of Matthias Lesch's 65 birthday. Bonn, Germany.
- ◇ DECEMBER 2025: NCG Colloquim II. Nijmegen, Netherlands.
- ◇ AUGUST 2025: NCG Day. Leiden, Netherlands.
- ◇ JUNE 2025: C^* -algebras, Coarse Geometry and Physics. Greifswald, Germany.
- ◇ APRIL 2025: Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Space-time. Marseille, France.
- ◇ OCTOBER 2024: Noncommutativity behind the Dunes. Delft, Netherlands.
- ◇ AUGUST 2024: OdenSeaG 2024. Odense, Denmark.
- ◇ MAY 2024: Leiden–Birmingham meeting. Leiden, Netherlands.
- ◇ APRIL 2024: Group Operator Algebras and Noncommutative Geometry. Marseille, France.
- ◇ NOVEMBER 2023: Leiden–Luxembourg PhD Away Day. Leiden, Netherlands.
- ◇ OCTOBER 2023: Autumn School on Large-scale Geometry. Göttingen, Germany.
- ◇ AUGUST 2023: YMC*A. Leuven, Belgium.
- ◇ MAY 2023: Noncommutative Geometry and Operator Algebras (NSeaG 2023). Bonn, Germany.
- ◇ MARCH 2023: Noncommutative Geometry and Representation Theory. Athens, Greece.
- ◇ MARCH 2022: Noncommutativity in the North. Gothenburg, Sweden.
- ◇ FEBRUARY 2022: Winter School on Foliations, Pseudodifferential Operators and Groupoids. Göttingen, Germany.

Organisation

- ◇ JANUARY 2023: [5th Conference of Settat on Operator Algebras and Applications](#).
Co-organisier, with Francesca Arici, Marcel de Jeu, Rachid El Harti and Dimitris Gerontogiannis.
- ◇ I organised several [Leiden Local NCG Seminars](#) and took [notes](#) for them.
 - SPRING 2022: KK-theory ([Notes](#)).
 - FALL 2022: Groupoid C^* -algebras ([Notes](#)).
 - FALL 2023: Noncommutative geometry of foliations ([Notes](#)).

RESEARCH VISITS

- ◇ DECEMBER 2025: Oberwolfach, Germany. Hosted by Dr. Jialong Deng.
- ◇ NOVEMBER 2025: Max Planck Institute for Mathematics, Germany. Hosted by Dr. Malte Leimbach.
- ◇ JULY 2025: Leibniz Universität Hannover, Germany. Hosted by Dr. Eske Ewert.
- ◇ JULY 2025: Georg-August-Universität Göttingen, Germany. Hosted by Prof. Dr. Ralf Meyer.
- ◇ JANUARY 2025: Universität Greifswald, Germany. Hosted by Prof. Dr. Matthias Ludewig.
- ◇ OCTOBER 2024: Peking University, China. Hosted by Dr. Guo Chuan Thiang.
- ◇ SEPTEMBER 2024: East China Normal University, China. Hosted by Prof. Dr. Hang Wang.
- ◇ SEPTEMBER 2024: Sichuan University, China. Hosted by Dr. Xingni Jiang and Dr. Yunfeng Shi.

TEACHING EXPERIENCE

Universiteit Leiden

- ◇ FALL 2024: Functional Analysis, Mastermath course. Teaching assistant.

- ◇ SPRING 2024: Operator Algebras, Mastermath course. Teaching assistant.
- ◇ FALL 2023: Functional Analysis, Mastermath course. Teaching assistant.
- ◇ SPRING 2023: Differentiable Manifolds II, undergraduate course. Teaching assistant.
- ◇ FALL 2022: Functional Analysis, Mastermath course. Teaching assistant.
- ◇ SPRING 2022: Differentiable Manifolds II, undergraduate course. Teaching assistant.
- ◇ FALL 2021: Functional Analysis, Mastermath course. Teaching assistant.